

network. We identified scalability as one concern. Suppose that when a mobile user moves from one network to another, the new foreign network advertises a specific route to the mobile user, and the old foreign network withdraws its route. Consider how routing information propagates in a distance-vector algorithm (particularly for the case of interdomain routing among networks that span the globe).

- a. Will other routers be able to route datagrams immediately to the new foreign network as soon as the foreign network begins advertising its route?
  - b. Is it possible for different routers to believe that different foreign networks contain the mobile user?
  - c. Discuss the timescale over which other routers in the network will eventually learn the path to the mobile users.
- P12. Suppose the correspondent in Figure 6.22 were mobile. Sketch the additional network-layer infrastructure that would be needed to route the datagram from the original mobile user to the (now mobile) correspondent. Show the structure of the datagram(s) between the original mobile user and the (now mobile) correspondent, as in Figure 6.23.
- P13. In mobile IP, what effect will mobility have on end-to-end delays of datagrams between the source and destination?
- P14. Consider the chaining example discussed at the end of Section 6.7.2. Suppose a mobile user visits foreign networks A, B, and C, and that a correspondent begins a connection to the mobile user when it is resident in foreign network A. List the sequence of messages between foreign agents, and between foreign agents and the home agent as the mobile user moves from network A to network B to network C. Next, suppose chaining is not performed, and the correspondent (as well as the home agent) must be explicitly notified of the changes in the mobile user's care-of address. List the sequence of messages that would need to be exchanged in this second scenario.
- P15. Consider two mobile nodes in a foreign network having a foreign agent. Is it possible for the two mobile nodes to use the same care-of address in mobile IP? Explain your answer.
- P16. In our discussion of how the VLR updated the HLR with information about the mobile's current location, what are the advantages and disadvantages of providing the MSRN as opposed to the address of the VLR to the HLR?

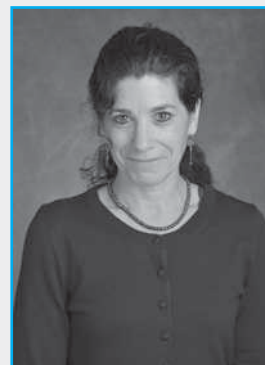


## Wireshark Lab

At the companion Web site for this textbook, <http://www.awl.com/kurose-ross>, you'll find a Wireshark lab for this chapter that captures and studies the 802.11 frames exchanged between a wireless laptop and an access point.

## Deborah Estrin

Deborah Estrin is Professor of Computer Science at UCLA, the Jon Postel Chair in Computer Networks, Director of the Center for Embedded Networked Sensing (CENS), and co-founder of the non-profit [openmhealth.org](http://openmhealth.org). She received her Ph.D. (1985) in Computer Science from M.I.T., and her B.S. (1980) from UC Berkeley. Estrin's early research focused on the design of network protocols, including multicast and inter-domain routing. In 2002 Estrin founded the NSF-funded Science and Technology Center, CENS (<http://cens.ucla.edu>), to develop and explore environmental monitoring technologies and applications. Currently Estrin and collaborators are developing **participatory sensing** systems, leveraging the programmability, proximity, and pervasiveness of mobile phones; the primary deployment contexts are mobile health (<http://openmhealth.org>), community data gathering, and STEM education (<http://mobilizingcs.org>). Professor Estrin is an elected member of the American Academy of Arts and Sciences (2007) and the National Academy of Engineering (2009). She is a fellow of the IEEE, ACM, and AAAS. She was selected as the first ACM-W Athena Lecturer (2006), awarded the Anita Borg Institute's Women of Vision Award for Innovation (2007), inducted into the WITI hall of fame (2008) and awarded Doctor Honoris Causa from EPFL (2008) and Uppsala University (2011).



**Please describe a few of the most exciting projects you have worked on during your career. What were the biggest challenges?**

In the mid-90s at USC and ISI, I had the great fortune to work with the likes of Steve Deering, Mark Handley, and Van Jacobson on the design of multicast routing protocols (in particular, PIM). I tried to carry many of the architectural design lessons from multicast into the design of ecological monitoring arrays, where for the first time I really began to take applications and multidisciplinary research seriously. That interest in jointly innovating in the social and technological space is what interests me so much about my latest area of research, mobile health. The challenges in these projects were as diverse as the problem domains, but what they all had in common was the need to keep our eyes open to whether we had the problem definition right as we iterated between design and deployment, prototype and pilot. None of them were problems that could be solved analytically, with simulation or even in constructed laboratory experiments. They all challenged our ability to retain